组会报告

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Motivation

Mao, W., Wang, W., Dou, Z. et al. Fire Recognition Based On Multi-Channel Convolutional Neural Network. Fire Technol 54, 531–554 (2018).

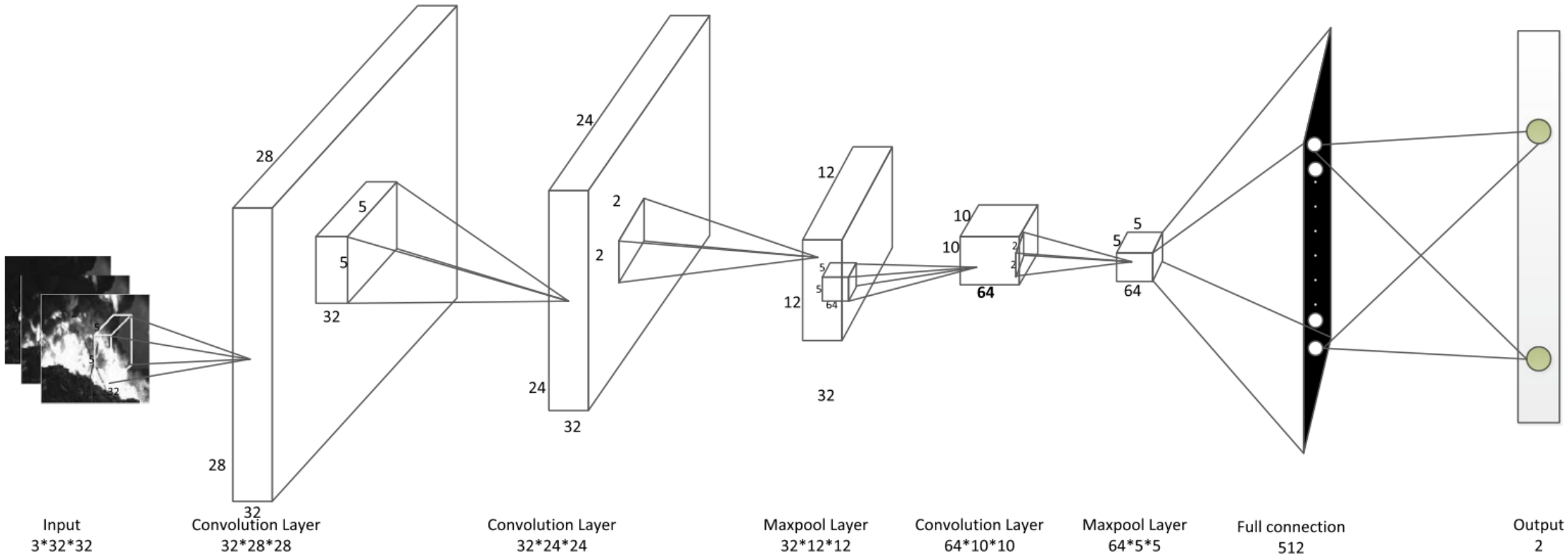
火灾识别很重要。

传统方法：基于烟雾传感器、热释红外火焰检测器或紫外线火焰检测器的物理信号。
缺点：极度依赖外部环境，通常只在小区域内工作，覆盖大范围区域需要大量传感器。

模式识别：先提取特征，然后训练识别模型（SVM或ANN）。
缺点：
1.需要手工提取特征，极大依赖先验知识。
2.大多数ANN和SVM是浅层网络，容量不够，在复杂背景下不能有效识别火焰。
3.大多数识别方法用于固定场景和单个火焰类型。

Architecture

Mao, W., Wang, W., Dou, Z. et al. Fire Recognition Based On Multi-Channel Convolutional Neural Network. Fire Technol 54, 531–554 (2018).



网络架构

V-SIFT

Mao, W., Wang, W., Dou, Z. et al. Fire Recognition Based On Multi-Channel Convolutional Neural Network. Fire Technol 54, 531–554 (2018).

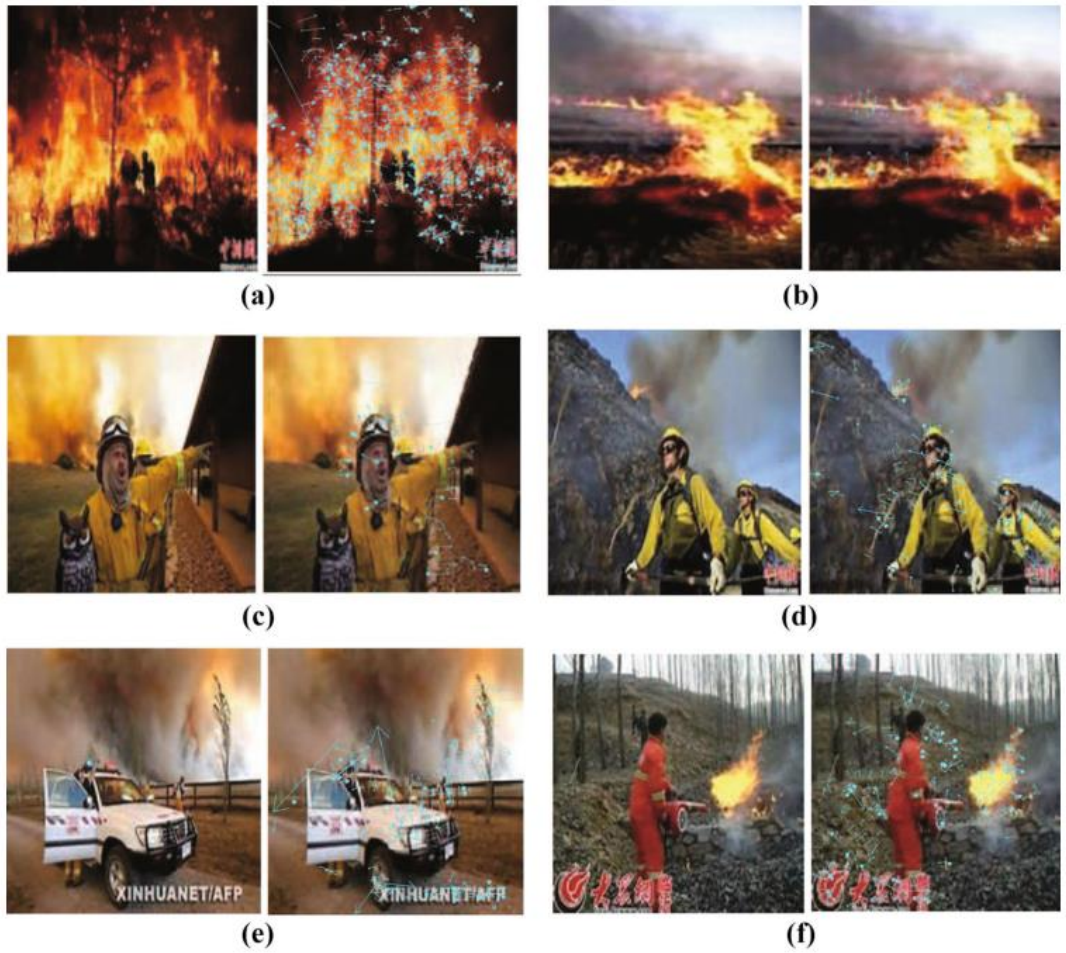


Figure 7. Feature points obtained by V-SIFT for six sample fire images. In each image the left column is original image and the right column gives the extracted feature map from V-SIFT respectively. The blue lines represent key points and their direction vectors (Color figure online).

a,b 识别效果较好，c-f 复杂背景识别准确率下降。

Feature maps

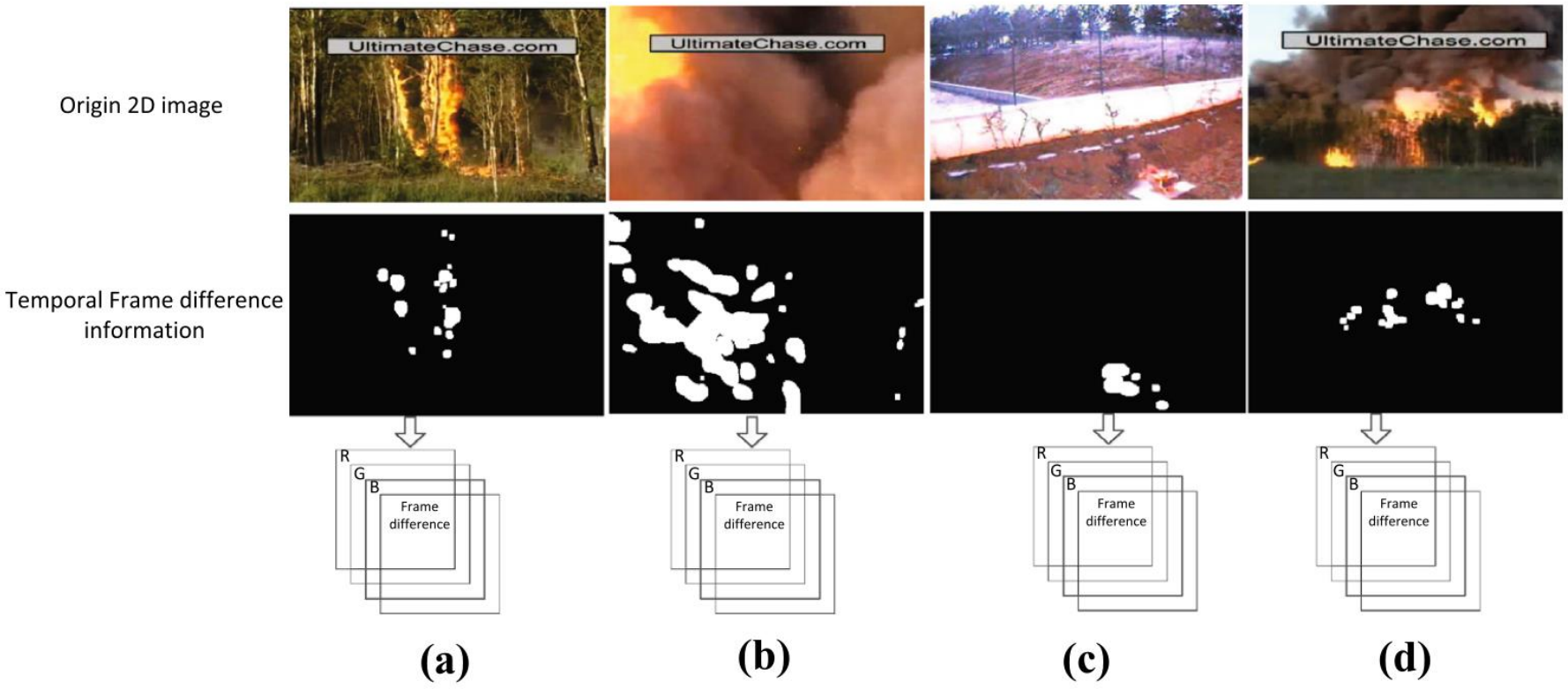
Mao, W., Wang, W., Dou, Z. et al. Fire Recognition Based On Multi-Channel Convolutional Neural Network. Fire Technol 54, 531–554 (2018).

Origin image　Feature map1　Feature map2　Feature map3

网络特征图

Temporal

Mao, W., Wang, W., Dou, Z. et al. Fire Recognition Based On Multi-Channel Convolutional Neural Network. Fire Technol 54, 531–554 (2018).



Figrue 11. The fourth channel input of MCCNN using temporal frame difference information.

从数据集的视频中，计算连续帧的帧差异，作为时间信息输入 MCCNN。帧差异可以大致描绘火焰轮廓，但也容易受到烟和树等噪声影响。

Experiments

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Table 2
Comparative Results of Six Methods

No	Method	Train time (s)	Test time (s)	Train set accuracy (%)	Test set accuracy (%)	Recall (%)	Precision (%)	F1 score
1	DNN	410.6	0.23	84.1	96.93	97.44	96.51	0.9697
2	V-SIFT	**43.84**	9.21	97.31	96.14	96.34	96.29	0.9631
3	SAE	1522.5	**0.2**	99.08	97.22	96.96	97.52	0.9724
4	DBN	2455.9	0.4	99.03	97	96.78	97.25	0.9701
5	**MCCNN**	15000 (GPU)	8.13	**99.86**	**98.06**	**97.62**	**98.53**	**0.9805**
6	Temporal MCCNN	24015 (GPU)	10.15	96.01	92.54	84.23	71.25	0.7720

The bold values mean the best result in terms of each evaluation index

Temporal MCCNN 效果不好，猜测原因：

1. 火灾视频缺少多样性，帧差异信息增加相同数据的重复
2. 深度学习模型对重复数据相对敏感，使得模型偏向部分数据，引起过拟合。
3. 帧差异信息也可能不适合作为网络输入。